DSA0605 – Data Handling and Visualization

CO2-AT2- Visualization Development Exercise

Name: P. Siddardha

Reg.no: 192424271

Q1. Boxplots – Customer Waiting Times Across Four Ride-Hailing Zones

Aim: To simulate waiting times for four city zones and compare their distributions while identifying potential outliers.

R Program

```
# Q1: Boxplots of customer waiting times
library(ggplot2)

# Generate simulated waiting-time data
set.seed(123)

wait_data <- data.frame(
  Zone = rep(c("North", "South", "East", "West"), each = 30),
  WaitTime = c(
    rnorm(30, 5, 1.5),
    rnorm(30, 7, 2),
    rnorm(30, 6, 1.8),
    rnorm(30, 9, 2.5)
  )
)

# Insert unusual values to demonstrate outliers
wait_data$WaitTime[5] <- 18
wait_data$WaitTime[95] <- 25

# Save the dataset
write.csv(wait_data, "Q1_Waiting_Times.csv", row.names = FALSE)

# Construct the boxplot
p1 <- ggplot(wait_data, aes(x = Zone, y = WaitTime, fill = Zone)) +
  geom_boxplot() +
  labs(
    title = "Customer Waiting Times Across Four City Zones",
    x = "Ride-Hailing City Zone",
    y = "Waiting Time (Minutes)",
    fill = "Zone"
  ) +
  scale_fill_brewer(palette = "Set2") +
```

```
  theme_minimal()

print(p1)

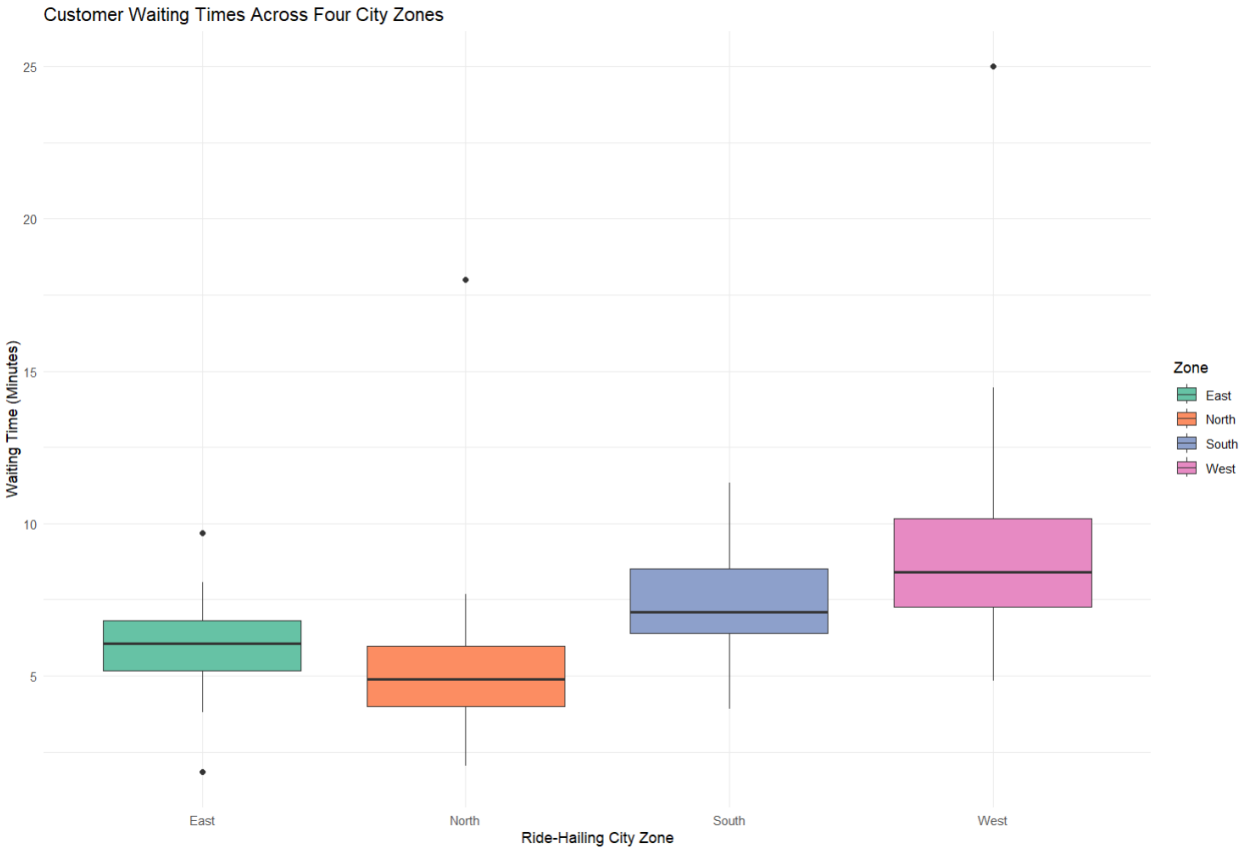
# Save the chart at 300 DPI
ggsave("Q1_Boxplot.png", p1, width = 8, height = 6, dpi = 300)
```

Dataset Customer Waiting Times Across Four Ride-Hailing Zones

Zone	WaitTime (Minutes)
North	18.0
North	4.5
North	5.2
North	4.8
North	5.5
South	7.2
South	6.5
South	8.1
South	5.9
South	7.5
East	5.8
East	6.2
East	4.9
East	7.1
East	5.5
West	8.5
West	9.2
West	7.8
West	10.1
West	25.0

Interpretation:

- The median line represents the typical waiting time in each zone.
- The box shows the interquartile range, while the whiskers indicate the spread of values.
- Points beyond the whiskers represent potential outliers.
- The West zone has a higher simulated average waiting time than the other zones.



Conclusion: Boxplots effectively compare waiting-time distributions and identify unusual values across multiple groups.

Q2. Bar Chart – EV Charging Sessions at Five Stations

Aim: To compare charging sessions completed at five electric-vehicle charging stations during the previous month and export the chart as a 300-DPI PNG.

R Program

```
# Q2: EV charging sessions
library(ggplot2)

# Create the dataset
charging <- data.frame(
  Station = c("Station A", "Station B", "Station C",
              "Station D", "Station E"),
  Sessions = c(420, 560, 480, 650, 390)
)
```

```
# Save the dataset
write.csv(charging, "Q2_Charging_Sessions.csv", row.names = FALSE)

# Construct the bar chart
p2 <- ggplot(charging, aes(x = Station, y = Sessions, fill = Station)) +
  geom_col() +
  labs(
    title = "EV Charging Sessions at Five Stations",
    x = "Charging Station",
    y = "Number of Charging Sessions",
    fill = "Station"
  ) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()

print(p2)

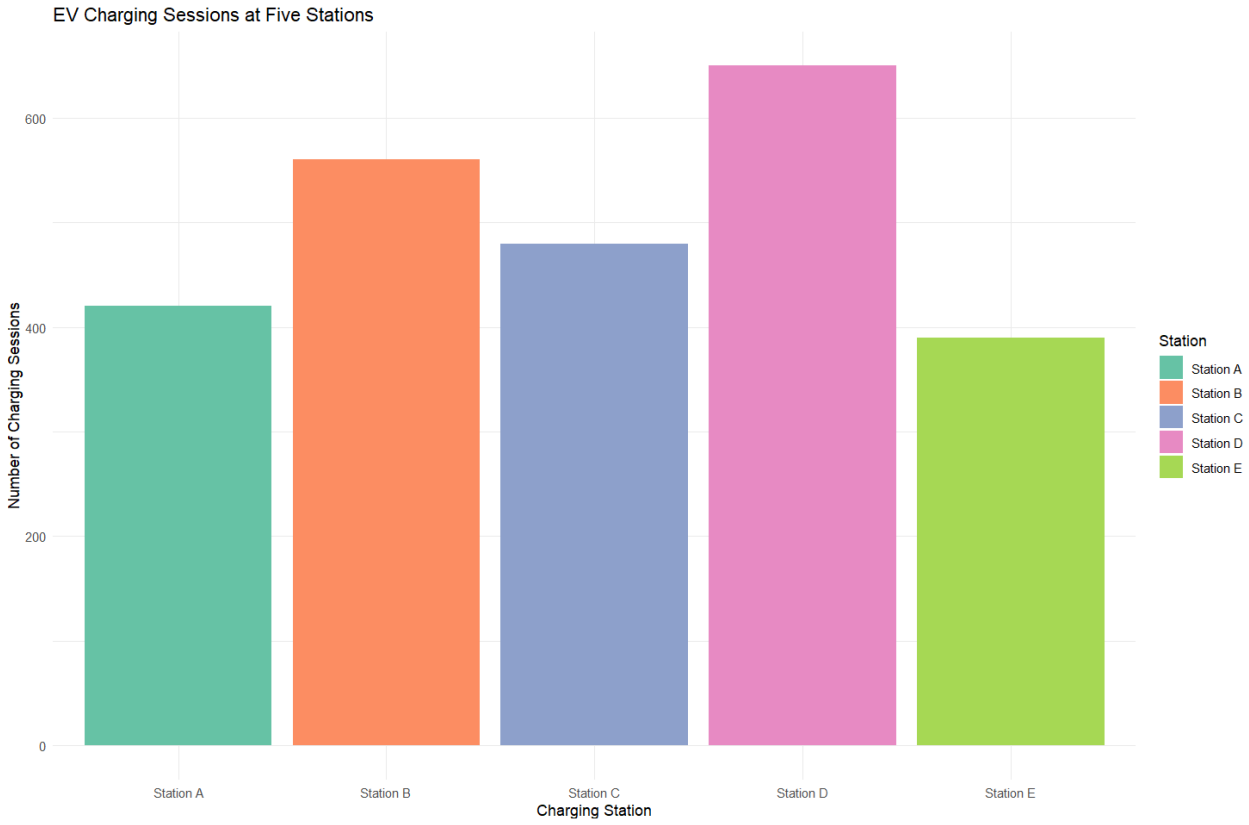
# Export the chart as a high-resolution PNG
ggsave(
  filename = "Q2_EV_Charging.png",
  plot = p2,
  width = 8,
  height = 6,
  units = "in",
  dpi = 300
)
```

Dataset EV Charging Sessions at Five Stations

Station	Sessions
Station A	420
Station B	560
Station C	480
Station D	650
Station E	390

Interpretation:

- Station D completed the highest number of sessions (650).
- Station E completed the lowest number of sessions (390).
- The bar heights make comparisons between stations easy.



Conclusion: A bar chart is suitable for comparing numerical counts across categories. The ggsave() function exports the chart at 300 DPI.

Q3. Histogram – Flight Delay Distribution for 38 Domestic Flights

Aim: To generate flight-delay durations for 38 flights and display their distribution using a function argument to control the number of bins.

R Program

```
# Q3: Flight-delay distribution
library(ggplot2)

# Generate delay data for 38 flights
set.seed(123)

flight_data <- data.frame(
  Flight = 1:38,
  Delay = round(pmax(0, rnorm(38, mean = 20, sd = 12)), 1)
)
```

```
# Save the dataset
write.csv(flight_data, "Q3_Flight_Delays.csv", row.names = FALSE)

# Function to plot a histogram with a customizable bin count
plot_delay <- function(number_of_bins = 8) {
  ggplot(flight_data, aes(x = Delay)) +
    geom_histogram(
      bins = number_of_bins,
      fill = "skyblue",
      color = "black"
    ) +
    labs(
      title = "Distribution of Domestic Flight Delays",
      x = "Flight Delay (Minutes)",
      y = "Number of Flights"
    ) +
    theme_minimal()
}

# Display histogram using 8 bins
p3 <- plot_delay(number_of_bins = 8)
print(p3)

# Save the chart
ggsave("Q3_Flight_Delays.png", p3, width = 8, height = 6, dpi = 300)

# Change the bin count when needed:
# print(plot_delay(number_of_bins = 12))
```

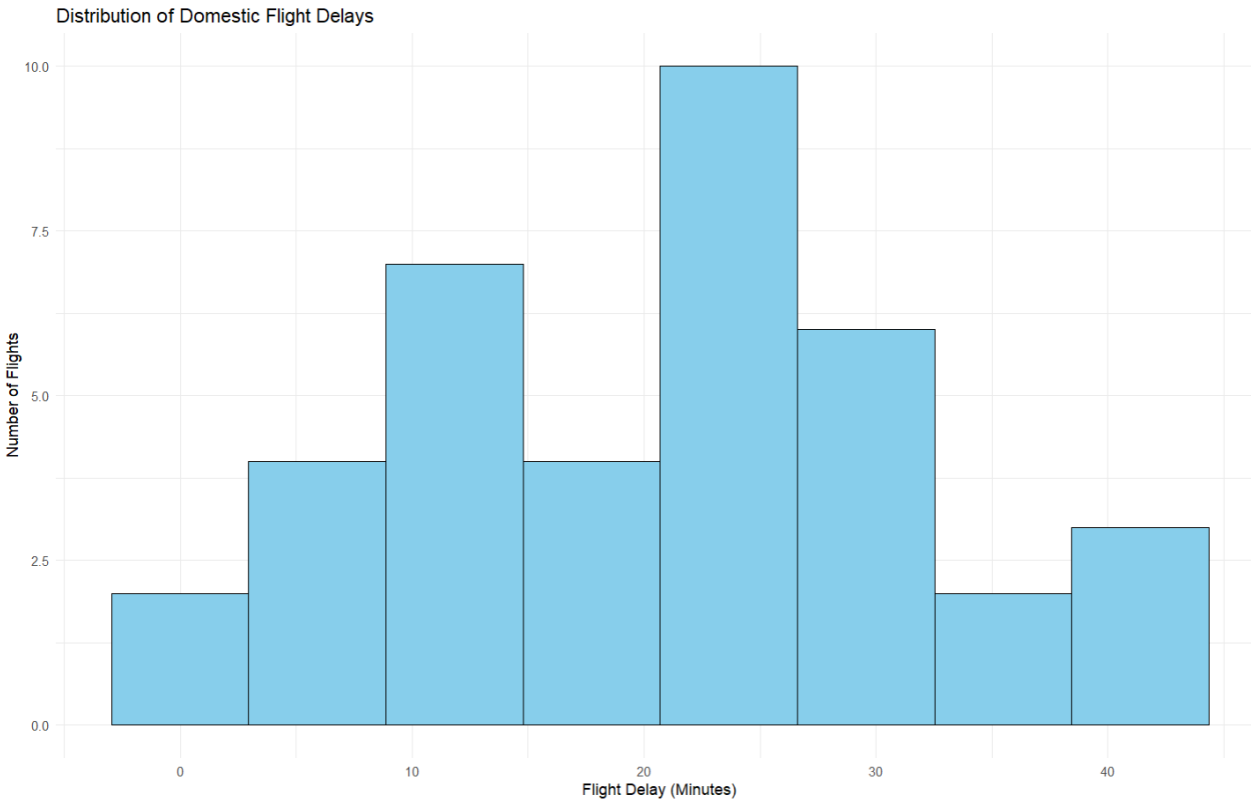
Dataset Flight Delays for 38 Domestic Flights

The following is a sample dataset. The R program generates the full 38 observations using set.seed(123).

Flight	Delay (Minutes)
1	13.3
2	17.2
3	38.7
4	20.8
5	21.6
6	40.6
7	25.5

8	4.8

Interpretation:
- The histogram groups the 38 flight delays into intervals.
- Taller bars indicate more flights within a particular delay interval.
- Increasing the number of bins provides more detail; fewer bins provide a broader overview.



Conclusion: A histogram is appropriate for visualizing the distribution of numerical data. The number of bins argument controls the chart's level of detail.

Q4. Pie Chart – Proportion of Wildlife Sightings Across Four Species Groups

Aim: To calculate the proportions of wildlife sightings from raw counts and visualize the percentage contribution of four species groups.

R Program

```
# Q4: Proportions of wildlife sightings
library(ggplot2)
```

```
# Create raw sighting-count data
sightings <- data.frame(
  Species = c("Mammals", "Birds", "Reptiles", "Amphibians"),
  Count = c(250, 400, 150, 200)
)

# Calculate proportions and percentages
sightings$Proportion <- sightings$Count / sum(sightings$Count)
sightings$Percentage <- sightings$Proportion * 100

# Display calculated values
print(sightings)

# Save the dataset
write.csv(sightings, "Q4_Wildlife_Sightings.csv", row.names = FALSE)

# Construct the pie chart
p4 <- ggplot(sightings, aes(x = "", y = Proportion, fill = Species)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  geom_text(
    aes(label = paste0(round(Percentage, 1), "%")),
    position = position_stack(vjust = 0.5)
  ) +
  labs(
    title = "Proportion of Wildlife Sightings by Species",
    fill = "Species Group"
  ) +
  theme_void()

print(p4)

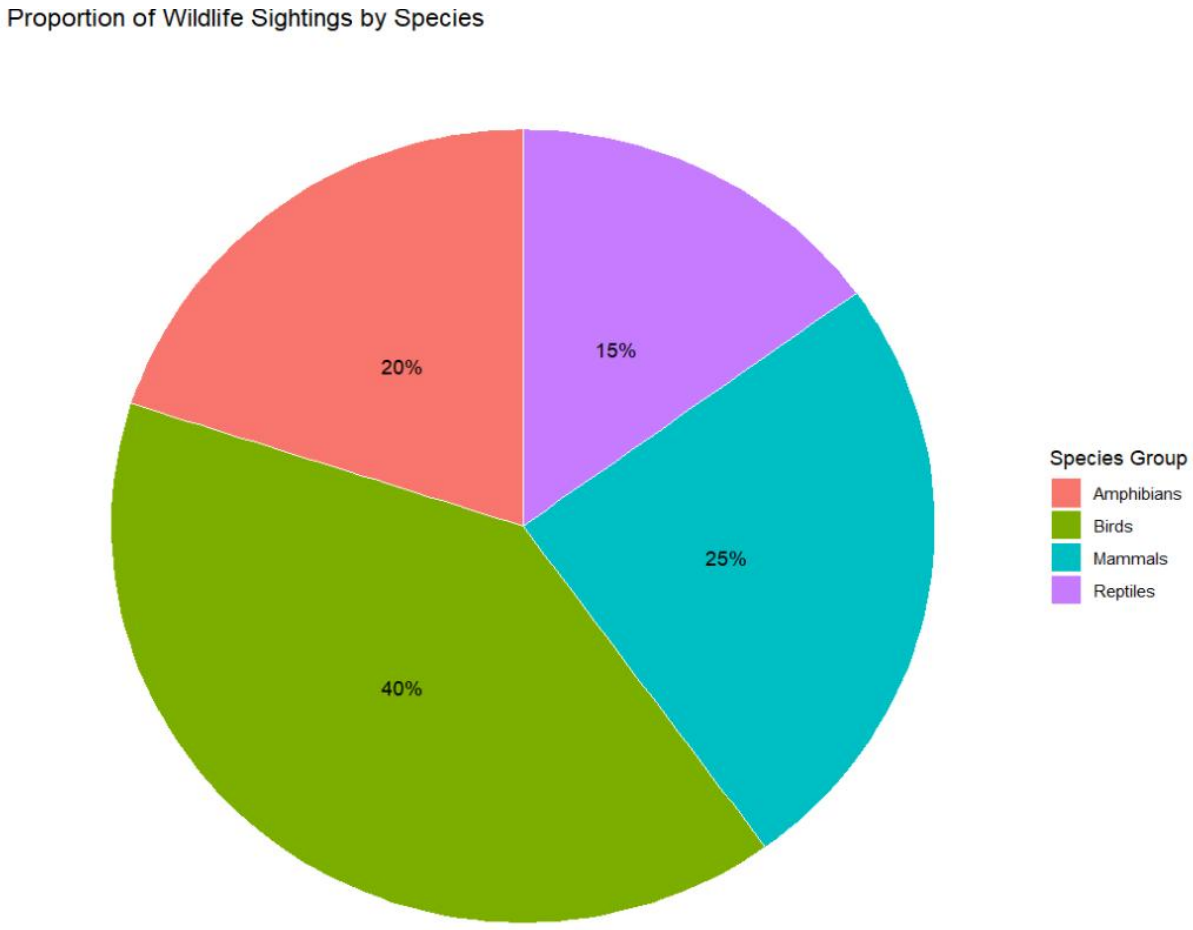
# Save the chart
ggsave("Q4_Wildlife_Pie_Chart.png", p4, width = 8, height = 6, dpi = 300)
```

Dataset Wildlife Sanctuary Animal Sightings

Species	Count	Proportion	Percentage
Mammals	250	0.25	25%
Birds	400	0.40	40%
Reptiles	150	0.15	15%

Amphibians	200	0.20	20%
Total	1000	1.00	100%

Interpretation:
- Birds account for 40% of recorded sightings.
- Mammals account for 25%.
- Amphibians account for 20%.
- Reptiles account for 15%.



Conclusion: A pie chart effectively shows how the total 1,000 sightings are divided among four species groups. All percentages add up to 100%.

Q5. Scatter Plot – Brewery Advertising Spend and Monthly Keg Sales

Aim: To simulate advertising expenditure and keg sales over 20 months and examine their relationship using a scatter plot with a fitted trend line.

R Program

```
# Q5: Advertising spend versus keg sales
library(ggplot2)

# Simulate data for 20 months
set.seed(123)

brewery <- data.frame(
  Month = 1:20,
  AdvertisingSpend = round(runif(20, 5000, 25000), 0)
)

# Generate sales with a positive relationship to advertising
brewery$KegSales <- round(
  80 + 0.012 * brewery$AdvertisingSpend + rnorm(20, 0, 20)
)

# Save the dataset
write.csv(brewery, "Q5_Brewery_Data.csv", row.names = FALSE)

# Construct scatter plot and fitted linear trend line
p5 <- ggplot(
  brewery,
  aes(x = AdvertisingSpend, y = KegSales)
) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Relationship Between Advertising Spend and Keg Sales",
    x = "Monthly Advertising Spend (₹)",
    y = "Monthly Keg Sales (Number of Kegs)"
  ) +
  scale_x_continuous(labels = function(x) {
    paste0("₹", format(x, big.mark = ",", scientific = FALSE))
  }) +
  theme_minimal()

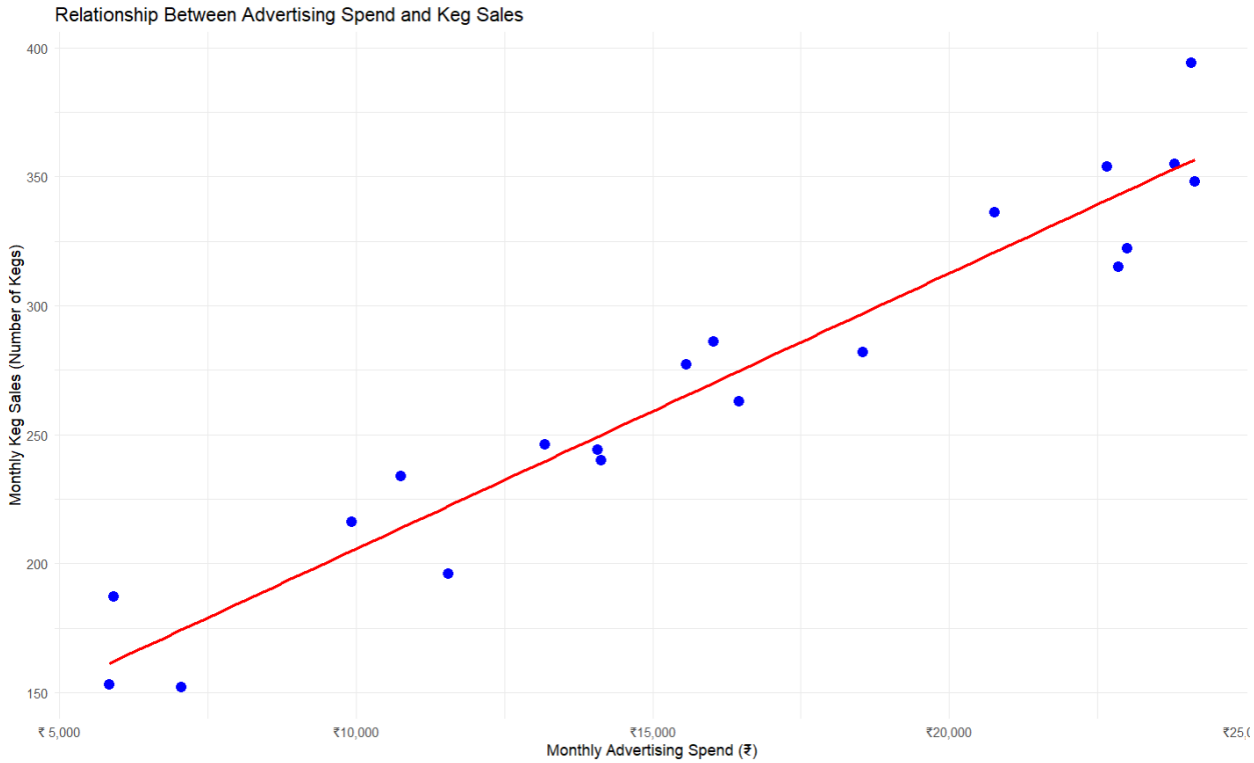
print(p5)
```

Save the chart
ggsave("Q5_Brewery_Scatter.png", p5, width = 8, height = 6, dpi = 300)

Dataset Brewery Advertising Spend and Monthly Keg Sales

Month	Advertising Spend (₹)	Keg Sales
1	10,753	196
2	20,766	320
3	13,180	220
4	22,660	345
5	23,814	350
6	5,911	160

Interpretation:
- Each point represents one month's advertising spend and keg sales.
- The red line represents the fitted linear trend.
- The upward trend indicates a positive relationship in the simulated data.
- This association does not prove that advertising alone causes sales to increase.



Conclusion: A scatter plot with a fitted regression line is suitable for examining the relationship between two numerical variables.